AI IN SPACE AND ORBITAL OPERATIONS - VOLUME 6

The AI Ground Systems and Mission Software Playbook

How AI runs the ground half of every mission - stations, pipelines, planning and code - without ever mistaking a green console for a trustworthy ground system.



Gates GS0-GS5 | Eight templates | 20 golden rules

A green console is not a trustworthy ground system. Ground and mission software AI is licensed per mission, per station network, per software release and per pass - never in general, and never on a test passed once in a lab.

The claim ladder for this volume: coded, tested, integrated, rehearsed, transferable. A system earns the rung it can prove under realistic load, realistic failure and a realistic clock - not the rung its demo reached.

Every spacecraft is flown by software on the ground; the question is how much of it you trust, and on what evidence

The ground segment is where missions are actually flown. Antennas are rented, passes are minutes long, and between passes the mission exists only as ground software: schedules, ephemerides, command loads, telemetry archives and the people watching them. AI enters this segment with attractive claims - automated pass operations, predictive station maintenance, planning copilots, anomaly assistants - and one structural danger: ground software fails quietly, at scale, and usually at the worst pass.

- Name the missions, station networks and software releases in scope - a licence for one mission's ground chain says nothing about another's.
- State the claim on the ladder: coded, tested, integrated, rehearsed or transferable - and the evidence that would move it.
- List the standing refusals on page one: no unverified command path, no unrehearsed release, no single point of ground failure hidden by automation.
- Price the downside in passes lost and spacecraft risk, not in engineering hours.

This volume treats the ground system as a flight system that happens to sit on Earth. It gets a safety case, a release discipline, a rehearsal regime and a licence - per mission, per station network, per release and per pass. Anything less is a console with good graphics.

Rule: if a ground change could lose a pass, corrupt a command or mislead a controller, it is a flight change - and it is reviewed, tested and rehearsed like one.

EXECUTIVE LENS

The ground segment is not overhead; it is half the spacecraft, and it is the half you can still patch.

MISSION LENS

A licence names missions and releases; 'the ground system is AI-ready' names nothing.

OPERATIONS LENS

Controllers trust what rehearsals proved, not what a release note promised.

Ten stages from station scheduling to software release, graded stage by stage



Figure 1 - One pass flowing through station, telemetry, flight dynamics, planning and verification, each stage on its own time plane.

Grade each stage separately. A mature telemetry pipeline does not redeem an unrehearsed release process, and an excellent planning tool does not excuse a command path with no independent check.

- 1. Station scheduling and network management. 2. Link acquisition and RF chain. 3. Telemetry ingest, decommutation and display. 4. Flight dynamics: orbit determination and prediction. 5. Mission planning and pass planning.
- 6. Command generation and load building. 7. Command verification and uplink authorisation. 8. Payload data reception, processing and distribution. 9. Archive, replay and provenance. 10. Software development, test and release for all of the above.

Stages 6 and 7 carry irreversible risk and stay under the tightest licence in this volume. Stages 3, 4, 5 and 8 are where AI earns its keep first: volume, pattern and prediction, with a human still able to see the raw stream underneath.

EXECUTIVE LENS

Map before you automate; most ground failures are seam failures between stages nobody owned.

MISSION LENS

Flight dynamics errors propagate into every plan and command built on them.

OPERATIONS LENS

The map is per mission; shared infrastructure inherits the strictest mission it serves.

Antennas are a market now; the licence is for the link you actually get, not the network on the brochure

Ground Station as a Service changed the economics: operators buy contacts by the minute from global networks instead of building stations. AI scheduling can raise utilisation and predict conflicts across hundreds of customers - and can also quietly deprioritise your emergency pass behind someone else's routine one, under rules you never read.

- Contract for priority classes in writing: routine, time-critical and emergency contacts, with pre-emption behaviour demonstrated, not described.
- Measure delivered link quality per pass - margin, data volume, acquisition time - and feed it back into scheduling decisions.
- Keep a sovereign or second-network fallback for emergency commanding; a single commercial network is a single point of failure with a logo.
- Audit the scheduler: whose passes moved, why, and what the AI optimised when yours lost out.

Station AI is licensed per network and per priority class. A scheduler proven on routine imaging passes has earned nothing about emergency commanding at 3 a.m. during a safe-mode event.

EXECUTIVE LENS

Cheap contacts that fail in an emergency are the most expensive contacts you will ever buy.

MISSION LENS

Priority is a tested behaviour, not a tier name in a price list.

OPERATIONS LENS

Know your pre-emption rank before the night you need it.

Telemetry, tracking and command data are the mission's bloodstream; AI may watch the flow, never silently reshape it

Telemetry pipelines decommute, calibrate, limit-check and archive the spacecraft's account of itself. AI adds real value here: smarter limit management, compression triage for constrained downlinks, and early pattern detection across thousands of parameters (Volume 3). The pipeline itself, however, is evidence infrastructure - and evidence infrastructure does not get to improvise.

- The raw stream is archived untouched before any AI processing; every derived product can be regenerated from raw.
- Calibration and decommutation changes are versioned releases with regression against golden passes, never configuration tweaks.
- AI triage ranks and annotates; it does not drop, smooth or re-scale telemetry values in the record.
- Pipeline latency and gaps are displayed as first-class health data - a silent pipeline is treated as a failed pipeline.

When the archive is trustworthy, every later argument - anomaly, insurance, inquiry - can be settled. When AI has quietly edited the record, nothing downstream can be believed, including the AI's own training data.

Rule: AI reads the bloodstream; it never transfuses it. The raw record is sacred, versioned and replayable.

EXECUTIVE LENS

Protect the raw archive first; every other ground investment depends on it.

MISSION LENS

A gap shown honestly is operations; a gap smoothed over is fiction.

OPERATIONS LENS

Golden-pass regression is cheap. Explaining a corrupted archive to an inquiry is not.

The uplink is the one ground product that cannot be recalled; it is checked by something dumber, separate and proud of it



Figure 2 - A command load passing through the amber independent verification ring before uplink.

Volumes 1 and 3 set the rule and this volume builds it in software: command loads are generated by capable systems and verified by an independent checker that shares no code, no model and no assumptions with the generator. AI may draft procedures, assemble loads and flag inconsistencies - the verification path stays deterministic, simple and separately owned.

- Generator and checker are built and maintained by different teams, on different stacks, from the same signed specification.
- The checker validates against hard limits - power, thermal, attitude, memory, forbidden states - not against the generator's intent.
- Every load carries its provenance: source plan, software versions, checker version, approver and expiry.
- A failed check stops the uplink by default; overriding a check is a named, logged, two-person act.

The moment the checker learns, adapts or shares a model with the planner, the ground system has one opinion stated twice. Independence is an architectural property, and it is tested by injecting generator faults and counting how many the checker catches.

EXECUTIVE LENS

Fund the checker like a flight component, because that is what it is.

MISSION LENS

Fault-injection scores for the checker belong in every release review.

OPERATIONS LENS

An override culture is visible in the logs long before it is visible in an incident.

Every plan, pass and command is built on an orbit prediction; its error bars are part of the product

Flight dynamics computes where spacecraft are and will be: orbit determination, manoeuvre reconstruction, event prediction and the ephemerides every other ground stage consumes. AI accelerates fitting, screens sensor residuals and forecasts conjunction geometry (Volume 5) - and can just as efficiently produce a precise, confident, wrong orbit if its residuals are never audited.

- Every ephemeris product ships with covariance and data-age metadata; consumers must refuse products that arrive without them.
- Orbit solutions are scored against withheld observations and independent solutions on a schedule, per object class.
- AI-assisted fitting is benchmarked against the classical estimator it assists; divergence is investigated, not averaged away.
- Manoeuvre reconstruction closes the loop: predicted versus flown is graded and fed back into model calibration.

Flight dynamics is a service with customers, and its service level is honesty about uncertainty. A ground system that distributes point predictions without error bars is distributing fiction with decimals.

EXECUTIVE LENS

Buy flight dynamics on scored accuracy, not on interface polish.

MISSION LENS

Withheld-observation scoring is the ground truth for the ground truth.

OPERATIONS LENS

When two solutions disagree, the disagreement itself is a product - surface it.

The plan is a promise the spacecraft, the stations and the crew all keep - or discover they cannot

Planning software turns mission intent into timelines: passes, payload activities, power and data budgets, ground tasks. AI planners explore far more of the schedule space than humans can and resolve routine conflicts well. Their characteristic failure is the feasible-looking plan that spends a margin nobody authorised - battery depth, recorder headroom, controller attention - because the margin was never encoded as a constraint.

- All margins are explicit, signed constraints in the planner, owned centrally (Volume 2); the planner may surface trade options, never spend margins silently.
- Every AI plan ships with its near-misses: the best rejected alternatives and what they would have cost or saved.
- Plans are graded as flown: scheduled versus executed, conflict causes, and margin consumption are reviewed weekly.
- Contingency capacity is planned, not hoped for: empty slots and reserve passes survive the optimiser (Volume 3).

A planner is licensed per mission and per planning horizon. The licence question is never 'does it produce plans?' but 'when its plan fails, does it fail visibly, early and inside the margins?'

EXECUTIVE LENS

Judge planners on as-flown grading, not on optimisation scores.

MISSION LENS

Near-miss transparency is what makes an AI plan reviewable in the time available.

OPERATIONS LENS

Reserve capacity the optimiser can spend is not reserve capacity.

Flight software and ground software are two halves of one machine, separated by a radio link and a release calendar

The spacecraft runs flight software; the ground runs its twin, its simulator, its command database and its displays. They evolve on different calendars and fail as a couple: a ground database one version ahead of the flight image has caused more real anomalies than most algorithms ever will. AI assistants that generate procedures, patches or database updates must respect the pairing as a hard constraint.

- A single configuration register pairs every flight image with its ground database, simulator and procedure versions; mismatches alarm before the pass, not during it.
- Changes are assessed for cross-side impact by default: a ground-only change is a claim that must be argued, not assumed.
- AI-generated procedures are compiled and executed against the paired simulator before a human ever sees them as candidates.
- The register, not memory or a wiki, is what the pre-pass briefing reads.

Treat the pair as one system with one licence. The question at every gate is whether the combined machine - orbit half and ground half - has been tested as it will actually fly.

Rule: version the pair, test the pair, license the pair. A half-updated mission is an untested mission.

EXECUTIVE LENS

Configuration drift between flight and ground is a defect class of its own - count it.

MISSION LENS

Simulators earn their keep the day they refuse a procedure reality would have refused.

OPERATIONS LENS

The register is boring infrastructure that prevents exciting incidents.

Ground software is believed at the level it was rehearsed: with the real clock, the real team and a failure injected

Unit tests prove code; rehearsals prove missions. The ground chain earns its ladder rungs in environments that reproduce the operational truth: engineering-model spacecraft and flatsats, recorded and synthetic telemetry, the actual station interfaces, and controllers working real timelines with failures injected without warning.

- Every release passes golden-pass regression: a library of recorded passes replayed with expected outputs compared automatically.
- Flatsat and hardware-in-loop testing is mandatory for anything touching commanding, flight dynamics inputs or telemetry calibration.
- Rehearsals include the ugly cases: missed acquisition, corrupt frames, station handover mid-pass, a checker rejection at T-minus two minutes.
- Rehearsal findings are defects with owners and deadlines, tracked like flight anomalies - not workshop notes.

An unrehearsed release is an experiment conducted on a spacecraft. The rehearsal programme is therefore not a support activity; it is the licensing authority's evidence factory.

EXECUTIVE LENS

Fund rehearsal capacity before automation capacity; it is the bottleneck for every later claim.

MISSION LENS

Injected failures find the seams briefings never mention.

OPERATIONS LENS

A rehearsal that cannot fail anyone teaches nothing.

No hot patches during a pass, no silent updates ever: the ground changes on a calendar the mission approves

Modern software practice prizes continuous deployment; mission operations prizes the opposite virtue - nothing changes near a critical event, and everything that changed can be named. AI coding assistants raise the stakes: they produce plausible changes faster than review culture was built to absorb, including in tools that were never classified as mission software at all.

- Freeze windows are absolute: no releases to the operational chain inside a defined period before critical passes and events, with emergency change as a signed exception process.
- Every operational change ships with rollback tested, not described: the previous version is restored in rehearsal before the new one is approved.
- Shadow and canary deployment applies on the ground too: new versions run in parallel on live data, outputs compared, before they drive anything (Volume 3's fleet discipline, applied to code).
- AI-generated code follows the same review, test and provenance rules as human code, plus disclosure: generated sections are marked in the record.

The release register is the ground system's memory of itself. If you cannot answer 'what exactly was running during that pass?' in one query, the investigation after an anomaly starts with archaeology.

EXECUTIVE LENS

Freeze windows feel slow until the first incident they prevent; then they feel cheap.

MISSION LENS

Tested rollback is the difference between a bad release and a bad week.

OPERATIONS LENS

Ask the one-query question in audits; the answer predicts incident quality.

CHAPTER 11 - GROUND CYBERSECURITY

The ground segment is the attack surface with the uplink at the end of it; AI defends it and must never become its weakest account

An attacker who owns the ground system owns the spacecraft, without ever touching space. Ground networks concentrate credentials, command paths and archives in ordinary IT environments, increasingly cloud-hosted and vendor-operated. AI security tooling helps - anomaly detection on operator behaviour, log triage, configuration drift detection - and AI assistants with broad repository or console access become high-value targets themselves.

- Command paths are segmented, hardware-backed where possible, and never reachable from general corporate or vendor networks.
- AI assistants operate with least-privilege, per-task credentials; a coding or operations copilot never holds standing uplink authority.
- Security monitoring treats ground AI behaviour as a monitored subject: queries, retrievals and generated changes are logged and reviewable.
- Incident playbooks include the space-specific case: suspected command-path compromise means passes are flown manually or not at all until cleared.

Volume 8 treats space security whole; this chapter sets the ground floor: the path from a phished password to a radiated command must cross independent barriers, each of which assumes the others may already have fallen.

Rule: design the ground network for the day an AI assistant is the compromised account - because that is now a standard threat model, not a thought experiment.

EXECUTIVE LENS

Segmentation is unglamorous and decisive; test it with red teams, not diagrams.

MISSION LENS

Least privilege applies doubly to tireless, fast, confident assistants.

OPERATIONS LENS

Manual fallback is a security control; keep the skill and the paperwork alive.

The archive is the mission's long memory and its legal record; AI products inherit its discipline or inherit its doubts

Beyond the telemetry raw record (Chapter 4), the ground holds payload data, planning history, command logs, voice loops and the AI products built on all of them. Provenance discipline decides whether any of it can be trusted later: which sensor, which calibration, which software, which model, whose approval.

- Every data product carries a provenance chain to raw inputs and processing versions; products that cannot produce their chain are labelled unverified.
- Retention is designed per class with legal, anomaly-investigation and model-training needs separated - training consent is not archive consent.
- Replay capability is tested: pick a random historical pass quarterly and reproduce its products bit-for-bit from the archive.
- Access to archives is logged; bulk extraction for model training is a governed event, not a side effect of employment.

Missions outlive their teams. The archive, with its provenance intact, is how a question asked in year nine gets an honest answer about year two.

EXECUTIVE LENS

Provenance is cheapest at capture and impossible to retrofit honestly.

MISSION LENS

Quarterly replay drills find archive rot while it is still a maintenance task.

OPERATIONS LENS

Separate the archive's masters from every working copy AI tools can touch.

A copilot for controllers earns trust the way a junior controller does: supervised, graded and never alone on the console

Control-room AI drafts procedures, summarises passes, answers 'have we seen this signature before?', watches for slow trends and prepares handovers. Used well it returns controller attention to judgement. Used badly it becomes an unaccountable second voice in the loop - confident, tireless and wrong in ways that sound like experience.

- Assistance is graded like staffing: supervised use, then supervised autonomy on defined task classes, with error rates published per class.
- Every AI suggestion shown on console carries its basis - source procedure, prior cases, data window - in one click; basis-free advice is not displayed.
- Controllers can dismiss assistance per shift without penalty or paperwork; dismissal rates are monitored as a product-quality signal.
- The AI never speaks on the voice loop, never acknowledges a checklist item, and never closes a procedure step. Humans close steps.

The design target is a controller who is harder to fool and easier to help than before - including helpfully telling the AI to be quiet during the passes where quiet is the requirement.

EXECUTIVE LENS

Publish assistant error rates per task class; trust follows the published number.

MISSION LENS

Dismissal is data. Rising dismissal is a failing product, not a failing crew.

OPERATIONS LENS

Keep the voice loop human; some channels are about accountability, not bandwidth.

When the spacecraft misbehaves, the ground's memory is the first instrument; AI makes it searchable, not infallible

Anomaly response depends on retrieval: prior anomalies, similar signatures, the procedure that worked, the one that made it worse. AI knowledge systems over anomaly databases, flight notes and vendor documentation can cut search time from hours to minutes - and can also synthesise a plausible-sounding history that never happened, at the exact moment nobody has time to check.

- Retrieval is quoted, not paraphrased: answers cite the source document and passage, and the console links straight to it.
- Generated summaries of past anomalies are labelled as generated and are never the sole basis for a command decision.
- The anomaly database itself is curated data: closure quality, root-cause confidence and as-flown outcomes are reviewed, because the AI will inherit every shortcut in it.
- Search failure is a displayed state - 'no matching case found' is a valid, valuable answer the system must be willing to give.

In an anomaly, a fast wrong answer is worse than a slow honest one, because it consumes the team's one scarcest resource: coordinated attention. License anomaly AI on citation fidelity measured in drills, not on demo recall.

Rule: in the anomaly room, AI quotes or stays silent. Synthesis is for the calm days and the training syllabus.

EXECUTIVE LENS

Citation fidelity in drills is the licence metric; measure it before you need it.

MISSION LENS

Curate the anomaly base like flight data - the AI is only as honest as its sources.

OPERATIONS LENS

'No case found' said early saves the hours a false lead would have burned.

CCSDS frames, open interfaces and boring standards are what let a ground system outlive its vendors

Space ground systems interoperate through standards - CCSDS telemetry and command formats, cross-support transfer services, standard time and ephemeris products - or they do not interoperate at all. AI tooling built on proprietary internal formats deepens lock-in at model speed: every generated artefact in a private format is another brick in a wall around your own mission.

- Interfaces to stations, partners and payload customers use published standards or fully documented open specifications; exceptions are registered with exit plans.
- AI models consume and produce standard products where a standard exists, so a model can be replaced without rebuilding the chain.
- Conformance is tested against reference implementations and partner systems, not against the vendor's own test suite alone.
- Data portability is an acceptance criterion: the full mission archive exports in standard formats, demonstrated annually.

Standards look like overhead in year one and like freedom in year seven, when the vendor, the model or the strategy changes and the mission must not.

EXECUTIVE LENS

Test export annually; portability claimed but never exercised is a hope.

MISSION LENS

Standards conformance is a procurement gate, not a roadmap item.

OPERATIONS LENS

Every proprietary format accepted today is a migration project accepted for later.

Sharing a ground system across missions multiplies both the savings and the blast radius; licence the sharing, not just the system

Fleets and agencies share ground infrastructure across missions to cut cost per pass - and inherit coupled risk: one release, one network fault or one compromised credential can touch every mission at once. AI scheduling and automation across a shared estate amplifies the coupling unless isolation is designed and tested.

- Missions are isolated by default: separate credentials, data partitions and release rings; sharing is an explicit, reviewed exception per component.
- Release rings stagger across missions so no single deployment can affect the whole estate simultaneously (Volume 3's wave discipline).
- Shared-service incidents are accounted per mission: each mission's licence records exposure to common-cause events.
- A mission can exit to a degraded independent mode - fewer features, own chain - and the exit is rehearsed, not theorised.

The economics of sharing are real, and so is common-cause failure. The licence states both numbers side by side: saving per pass, and missions simultaneously at risk per shared component.

EXECUTIVE LENS

Price shared infrastructure with its blast radius printed next to the saving.

MISSION LENS

Staggered release rings are the cheapest common-cause insurance available.

OPERATIONS LENS

Rehearse the exit per mission; an unrehearsed exit is a dependency with extra steps.

The ground burns, floods and gets ransomwared on Earth schedules; the spacecraft keeps orbiting on its own

Ground resilience is tested by ordinary terrestrial events: site power loss, network cuts, cloud region failure, ransomware, a vendor outage during a critical pass. The spacecraft does not pause while the ground recovers, so recovery objectives are written in orbits and passes, not in business hours.

- Recovery time and recovery point objectives are defined per function in passes: commanding capability first, then planning, then payload processing.
- Failover to the backup operations chain is rehearsed on live shadow traffic and, on a schedule, for real passes.
- Degraded-mode operations are designed and staffed: reduced station network, manual planning, stored command capability for spacecraft safety.
- Backups of archives, configurations and credentials are isolated from the operational network and restoration is timed in drills.

AI adds a resilience duty of its own: models and assistants must fail safe when their infrastructure fails - visibly unavailable, never confidently improvising on stale context.

Rule: a failover that has only ever happened in a document has never happened. Rehearse it on real passes or do not claim it.

EXECUTIVE LENS

Write recovery objectives in passes; the spacecraft does not recognise business hours.

MISSION LENS

Degraded mode is a designed product with staffing, not an improvisation.

OPERATIONS LENS

Stale-context AI during an outage is a hazard; make unavailability loud.

Renting the ground by the hour is legitimate; renting away the ability to command is not

Cloud hosting brings elasticity for data processing and clean economics for bursty workloads, and it moves mission-critical functions into someone else's region, jurisdiction and incident queue. Sovereignty questions are operational, not rhetorical: whose law governs the archive, whose outage takes your pass, whose staff can reach the command path.

- Commanding and command verification never depend on a single commercial cloud region; the safety path survives a full provider outage.
- Data residency and legal exposure are mapped per data class, including telemetry that reveals capabilities and payload data under customer or national restriction.
- Exit cost is engineered, not discovered: standard formats (Chapter 15), documented configurations and an annual portability exercise.
- Cloud AI services used in the chain are inventoried with their data flows; mission data sent to an external model is a governed disclosure.

The test is simple to state and hard to pass: if the provider relationship ended in ninety days, could the mission still be commanded, planned and archived? If not, the cloud is not a supplier; it is a co-owner.

EXECUTIVE LENS

Keep the safety path provider-independent, even at the cost of elegance.

MISSION LENS

Inventory external model data flows like any other disclosure channel.

OPERATIONS LENS

Ninety-day exit is the sovereignty metric; measure it annually.

Consoles shape decisions: what the ground shows, hides and alarms decides what the team can catch

Ground software is the controllers' sensory system. AI-generated displays, summarised trends and smart alarm filtering decide what a human notices in an eight-minute pass. The failure mode is well documented across industries: automation that hides raw state until its own model fails, leaving the human to reconstruct the situation from nothing, under time pressure, at the worst moment.

- Raw state is always one action away from any AI summary; summaries link down to the parameters and plots they summarise.
- Alarm filtering is audited like Volume 3's triage: suppressed alarms are sampled weekly and every suppression rule has an owner and an expiry.
- Mode awareness is designed: the console shows which automation is active, on what data, with what confidence state - before it shows anything else.
- Displays are tested in rehearsal with the actual crews; a screen nobody can read at minute six of a bad pass fails acceptance however elegant it looked in review.

Good ground UX is a safety system. It is specified, tested and licensed with the same seriousness as the command checker, because in the pass, it is part of the command path.

EXECUTIVE LENS

One action to raw state, always - put it in the acceptance tests.

MISSION LENS

Suppressed-alarm sampling catches the filter learning the wrong lesson.

OPERATIONS LENS

Test screens at minute six of a bad rehearsal pass, not in a design review.

Ground software vendors sell platforms; you are buying behaviour under failure, priced per verified pass

Ground systems concentrate vendor risk: mission planning suites, station networks, cloud platforms, AI assistants, each with roadmap power over your operations. Contracts must buy evidence and exit, not just features: scored accuracy, release discipline, incident transparency and data portability, with remedies that bite.

- Service levels are written on operational outcomes - passes supported, products delivered with provenance, checker fault-injection scores - not on platform uptime alone.
- Vendor model and software changes enter your change control: notification windows, regression evidence and rollback rights are contractual (Volume 1's vendor discipline, applied to code).
- Incident disclosure timelines are in the contract, including near-misses in shared infrastructure that affected other customers.
- Source, configuration and data escrow arrangements are tested; an escrow never opened is a story, not a safeguard.

Judge vendors in rehearsals and incidents, where their behaviour is data, rather than in evaluations, where it is theatre. The contract's job is to make the rehearsal behaviour the everyday behaviour.

EXECUTIVE LENS

Buy scored behaviour and tested exit; features are the easy part.

MISSION LENS

Change notification windows for vendor models belong in the signed contract.

OPERATIONS LENS

Open the escrow once a year; verify it builds and runs.

Ground automation pays in passes flown well, not in consoles consolidated or licences resold

Count ground economics the series way, in three columns: hours and savings claimed, hours verifiably released from rosters and budgets, and hours banked. The unit is the verified pass: a contact executed as planned, with commanding verified, telemetry archived with provenance and products delivered - divided into the fully loaded ground cost, including rehearsal, verification, security and the fallback capability you must keep anyway.

- Fully loaded cost includes the independent checker, rehearsal programme, archive and manual fallback staffing - the costs automation proposals most often omit.
- Savings that come from cutting rehearsal, verification or archive quality are transfers from the risk account, and are labelled as such.
- Station-network and cloud unit costs are tracked per pass and per gigabyte with priority class separated; emergency capability is priced visibly, not averaged away.
- AI licence costs are counted per mission and release, including re-validation after every material model or software change.

Done honestly, the ground business case usually survives - passes are expensive and good automation genuinely helps. Done on vendor arithmetic, it survives only until the first lost pass reprices everything.

Rule: cost per verified pass, with rehearsal and verification inside the number. A saving bought by thinning either is a loan from the anomaly budget.

EXECUTIVE LENS

Put the fallback you must keep anyway inside the automation business case.

MISSION LENS

Emergency capability priced visibly gets defended in budget rounds; averaged away, it gets cut.

OPERATIONS LENS

Re-validation after change is a recurring cost of AI, not a surprise.

Anyone can halt the ground automation; restarting it takes evidence, a rehearsal and two signatures

The operating model binds the volume together: a ground systems authority owning the licence per mission and release, a configuration register that tells the truth, a rehearsal programme that manufactures evidence, and a halt culture in which stopping the automation is a normal professional act, not an accusation.

- Roles: ground systems authority, mission operations lead, independent verification owner, security owner and release manager - named people, with deputies, per mission.
- Every licence states its rung, scope, evidence, expiry and the events that demote it automatically: a failed rehearsal, a checker miss, an unplanned vendor change.
- Halts are drilled and debriefed without blame; restarts require the failure understood, the fix regressed and a rehearsal pass flown.
- The ground wall (Figure 3) shows licences, releases, rehearsals due and open findings per mission - the same wall the worked example audits.

Ground systems fail quietly and missions fail loudly. This operating model exists to reverse that: to make the ground's doubts loud, early and cheap, while they are still on Earth where everything can be fixed.

EXECUTIVE LENS

Name the authority per mission; shared accountability is no accountability in a pass.

MISSION LENS

Automatic demotion triggers remove the politics from bad news.

OPERATIONS LENS

Debrief halts without blame or the next halt will come one pass too late.

WORKED EXAMPLE - AURORA ORBITAL, PART 1

The audit: a ground modernisation claim meets its own release register

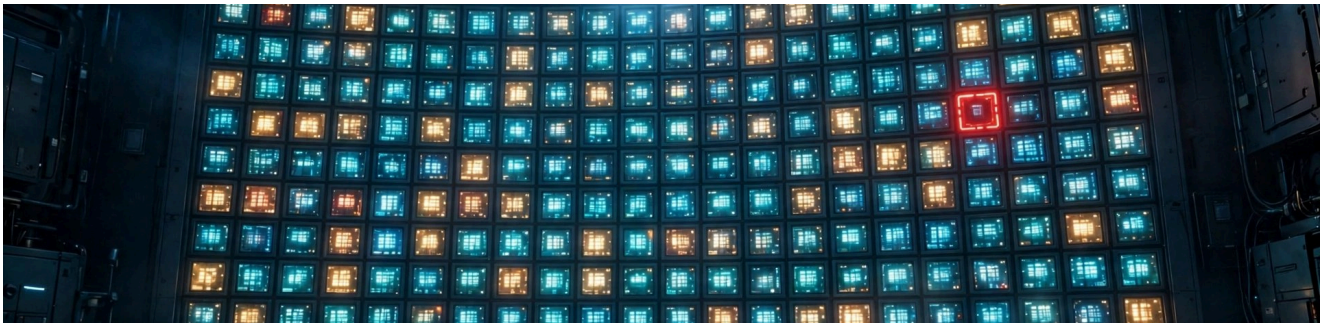


Figure 3 - The Aurora ground operations wall: licences, releases and rehearsals per mission, one cell held in red.

Fictional worked example. All organisations, figures and events are illustrative.

Aurora Orbital (Volumes 1-5) consolidated three mission ground chains onto one AI-assisted platform with a commercial station network and cloud-hosted processing. The vendor claimed a 47 percent ground-cost saving and 58,000 engineering and operator hours per year, with 'lights-out routine passes' across the fleet.

What the audit found

- Verified saving was 18 percent: 24,900 hours verifiably released and 15,200 banked, once rehearsal, verification and fallback staffing were costed back in.
- A ground database update shipped one version ahead of the flight image on two spacecraft; the configuration register existed but its pre-pass alarm had been muted during a console refresh - three passes were flown on a mismatched pair before a rehearsal caught the class of error (Chapter 8).
- The independent command checker caught 41 of 44 injected generator faults in release testing; the three misses shared a thermal-limit edge case and blocked the release until fixed and re-tested (Chapter 5).
- The station scheduler had quietly deprioritised Aurora's time-critical passes twice under network contention; priority behaviour was then demonstrated under test and written into the contract (Chapter 3).

None of this required new technology. It required the register to be believed, the checker to be funded and the scheduler to be audited - the unglamorous chapters of this volume, doing their work.

EXECUTIVE LENS

A platform claim is audited per mission and release, never accepted fleet-wide.

MISSION LENS

The muted alarm is the finding: safety signals that can be muted will be.

OPERATIONS LENS

Checker fault-injection scores decided a release date. That is the system working.

WORKED EXAMPLE - AURORA ORBITAL, PART 2

GS5 granted in part: the licence that survived was smaller and worth more

Fictional worked example, continued. All figures illustrative.

Function	Claimed	Licensed at GS4
Pass planning and station scheduling	Rung 5, fleet-wide	Rung 4 - rehearsed, per mission, priority classes contracted
Telemetry triage and archive	Rung 4	Rung 4 - raw record untouched, provenance complete
Command generation with independent check	Rung 4	Rung 4 - after checker re-test at 44 of 44 injected faults
Control-room and anomaly assistance	Rung 4	Rung 3 - supervised, citation-checked in drills
Lights-out routine passes	Rung 5	Refused - fallback staffing and rehearsal evidence did not support it
Unrehearsed vendor model updates	Standard practice	Refused - all changes through Aurora change control

Cost per verified pass settled at 0.83 against a 0.98 pre-platform baseline, fully loaded, with emergency commanding dual-networked and rehearsed. GS5 scale was granted in part: the platform extends to new missions one release ring at a time, each earning its own licence through rehearsal.

The vendor relationship improved under the new terms - scored behaviour, change windows, tested escrow - because the contract finally described the product Aurora actually needed: not a platform, but a ground system that could be trusted at 3 a.m.

EXECUTIVE LENS

Scale in part is the normal, healthy outcome of an honest gate.

MISSION LENS

Refusing lights-out passes kept the fallback real - and priced.

OPERATIONS LENS

A smaller true licence compounds; a grand false one detonates.

THE GROUND STANDARD

One page, to be read before any release reaches the operational chain

The Ground Standard. We license ground and mission software AI per mission, per station network, per software release and per pass. The raw telemetry record is never altered. Commands are verified by an independent checker that shares nothing with the generator. Flight and ground versions are licensed as a pair. Nothing unrehearsed reaches operations. Vendor changes enter our change control. Anyone can halt the automation; restart takes evidence, a rehearsal and two signatures.

- We claim only the rung we rehearsed: coded, tested, integrated, rehearsed, transferable.
- We freeze changes before critical passes, and we test rollback before we need it.
- We rehearse failover on real passes and price emergency capability visibly.
- We keep raw state one action away from every AI summary on every console.
- We count cost per verified pass, with rehearsal and verification inside the number.

Print it. Sign it per mission. A standard that lives only in this playbook governs nothing.

EXECUTIVE LENS

Sign it per mission; an unsigned standard is a poster.

MISSION LENS

The standard is short because the pass is short.

OPERATIONS LENS

Hold the standard in release reviews, where it costs something.

SIX FAILURE PATTERNS

How ground system programmes actually go wrong

Pattern	What happens	Antidote
1. The green console	Summaries hide raw state until the model fails mid-pass	Raw state one action away; suppressed-alarm audits
2. The mismatched pair	Ground database and flight image drift a version apart	Configuration register with unmutable pre-pass alarms
3. The helpful patch	An urgent fix bypasses rehearsal and becomes the incident	Freeze windows; emergency change as a signed exception
4. The single sky	One station network, cloud region or vendor holds the whole mission	Dual paths for commanding; ninety-day exit, exercised
5. The shared blast	One release or incident takes every mission on shared infrastructure	Isolation by default; staggered release rings
6. The edited record	AI processing quietly reshapes telemetry or archive data	Raw record sacred, versioned and replayable

Every pattern was survivable in rehearsal and expensive in operations. The patterns repeat because each begins as a reasonable shortcut under schedule pressure.

EXECUTIVE LENS

Patterns are cheaper to recognise in this table than in an anomaly report.

MISSION LENS

Schedule pressure is the common ancestor; govern it, not just its children.

OPERATIONS LENS

Rehearsal is where patterns go to be caught cheaply.

30 QUESTIONS

Asked at every gate, answered with evidence or not at all

- Which missions, networks and releases does this licence cover?
- What rung is claimed, and what rehearsal proves it?
- Is the raw telemetry record untouched and replayable?
- Who owns the independent checker, and what is its fault-injection score?
- Are flight and ground versions paired in the register right now?
- What changed since the last pass, anywhere in the chain?
- Is rollback tested for every component in this release?
- Which passes sit inside a freeze window?
- What priority does our emergency pass actually get, demonstrated when?
- Can we command if this cloud region or vendor fails tonight?
- What does the console hide, and how fast can a controller reach raw state?
- Which alarms were suppressed this week, and who sampled them?
- What is the AI assistant's published error rate per task class?
- Does anomaly support quote sources, and what is its drill citation score?
- Where does mission data flow to external models?
- Are interfaces standard, and was full archive export exercised this year?
- Which components are shared across missions, and what is the blast radius?
- When was failover last rehearsed on a real pass?
- Are recovery objectives written in passes?
- What demotes this licence automatically?
- Who can halt the automation, and when was a halt last drilled?
- What is the fully loaded cost per verified pass?
- Which savings depend on thinning rehearsal or verification?
- Are vendor change windows and incident disclosure in the signed contract?
- Has the escrow been opened and built this year?
- What is the ninety-day exit state for each critical supplier?
- Which procedures were AI-generated, and did they pass the paired simulator?
- How are AI assistant credentials scoped, and what happens if one is compromised?
- What evidence would make us reduce this licence tomorrow?
- Who signs the restart, and what rehearsal must precede it?

EXECUTIVE LENS

Thirty questions, one standard: evidence or it did not happen.

MISSION LENS

Ask them at gates, not after incidents.

OPERATIONS LENS

Any unanswered question is a rung the licence has not earned.

Six gates, each with the evidence that opens it

Gate	Meaning	Evidence required
GS0 - Frame	Scope the ground chain	Ground map per mission; standing refusals signed; register started
GS1 - Build	Systems coded and unit-tested	Paired versioning live; checker built independently; golden passes assembled
GS2 - Integrate	Chain works end to end	Flatsat and HIL results; fault-injection scores; failover design tested in shadow
GS3 - Rehearse	Operations proven under failure	Full rehearsal programme passed with crews; rollback and failover on real shadow passes
GS4 - License	Operational reliance at earned rungs	Per-mission licences; freeze and change control live; halt and restart drilled
GS5 - Scale	Extend missions and functions	Release-ring evidence per new mission; unit cost verified with rehearsal beside it

First 90 days

- **Days 1-30.** Build the ground map and configuration register; publish the standing refusals; freeze unrehearsed change.
- **Days 31-60.** Stand up the independent checker with fault injection; assemble golden passes; audit station priority behaviour.
- **Days 61-90.** Run full rehearsals including failover and a halt-and-restart; hold the first GS4 gate on evidence.

EXECUTIVE LENS

Gates open on rehearsals, not on demos.

MISSION LENS

The 90 days end with a shorter claim and a longer register.

OPERATIONS LENS

GS5 adds missions one ring at a time; it never widens a licence by itself.

What an auditor, a regulator or an inquiry will eventually ask for

The evidence pack, per mission

- Configuration register history: every flight-ground pairing, per pass.
- Release register with regression, rollback-test and approval records.
- Independent checker fault-injection scores per release.
- Rehearsal programme records and findings closure.
- Station network delivered-quality and priority-behaviour audits.
- Archive provenance and quarterly replay drill results.
- Failover and degraded-mode rehearsal evidence, timed.
- Vendor change notifications, incident disclosures and escrow test records.
- Halt, demotion and restart log with signatures.

Sector annex - where the emphasis moves

Sector	Emphasis
Constellation operators	Shared-infrastructure blast radius, release rings, pass economics at scale
Crewed and human spaceflight	Voice loop discipline, checker independence, freeze windows at maximum strictness
Deep-space missions	Archive and replay over decades, degraded modes, sparse-pass resilience (Volume 7)
Government and defence	Sovereign commanding path, segmented networks, protected data flows
Launch providers	Range and flight-safety ground chains, countdown freeze discipline
Small-satellite operators	GSaaS contract discipline, shared-service exit, minimum viable rehearsal

EXECUTIVE LENS

Assemble the pack before anyone asks; afterwards is too late and it shows.

MISSION LENS

The annex moves emphasis, never the standard.

OPERATIONS LENS

The register history is the evidence pack's spine.

EIGHT FILL-IN TEMPLATES

The licence, made paperwork

Template 1 - Ground Licence Card

Mission: ___ | Network / release: ___ | Rung: ___ | Scope: ___ | Expiry: ___ | Authority: ___

Template 2 - Configuration Pairing Record

Spacecraft / flight image: ___ | Ground database: ___ | Simulator: ___ | Procedures: ___ | Verified before pass: ___

Template 3 - Release and Rollback Record

Release: ___ | Golden-pass regression: ___ | Rollback tested: ___ | Freeze check: ___ | Approved by: ___

Template 4 - Checker Fault-Injection Log

Release: ___ | Faults injected: ___ | Caught: ___ | Misses and fixes: ___ | Checker version: ___ | Signed: ___

Template 5 - Pass Readiness Checklist

Pass: ___ | Pairing verified: ___ | Changes since last pass: ___ | Station priority: ___ | Fallback ready: ___ | Go / no-go: ___

Template 6 - Rehearsal Record

Scenario: ___ | Failures injected: ___ | Crew: ___ | Findings: ___ | Owners / due dates: ___

Template 7 - Failover Drill Log

Trigger: ___ | Functions failed over: ___ | Time in passes: ___ | Data loss: ___ | Findings: ___

Template 8 - Verified Pass Workbook

Period: ___ | Fully loaded ground cost: ___ | Verified passes: ___ | Cost per verified pass: ___ | Claimed / released / banked: ___

Licences, pairings, releases, checker scores, pass readiness, rehearsals, failovers and cost - each short enough to complete at 3 a.m., which is when several of them will be used. Paperwork from logs is evidence; from memory, fiction.

The volume in twenty lines, and the series in ten

- License per mission, per station network, per release and per pass.
- A green console is not a trustworthy ground system.
- The raw telemetry record is never altered by AI.
- Commands are checked by something independent, deterministic and dumber than the generator.
- Version the pair: flight and ground are licensed together.
- Nothing unrehearsed reaches the operational chain.
- No hot patches in a pass; rollback is tested, not described.
- Priority for emergency passes is demonstrated, not tier-named.
- Standards and export are exercised annually, or they do not exist.
- Shared infrastructure is priced with its blast radius beside the saving.
- Recovery objectives are written in passes, not business hours.
- The safety commanding path survives any single provider.
- AI assistance is graded per task class, with error rates published.
- In the anomaly room, AI quotes or stays silent.
- Raw state stays one action away from every summary.
- Vendor changes enter your change control, or they do not enter.
- Cost per verified pass, with rehearsal and verification inside the number.
- Anyone can halt the ground automation.
- Restart takes evidence, a rehearsal and two signatures.
- A smaller true licence beats a grand claim that folds.

AI in Space and Orbital Operations - the series

Vol	Title	Gates	Status
1	Space Strategy	SS0-SS5	Built
2	Mission Design and Autonomy	MD0-MD5	Built
3	Satellite and Constellation Operations	CO0-CO5	Built
4	Earth Observation and Data	EO0-EO5	Built
5	Conjunction, Debris and Orbital Safety	CD0-CD5	Built
6	Ground Systems and Mission Software	GS0-GS5	Built - this volume
7	Lunar and Deep-Space Operations	LD0-LD5	Next
8	Space Security and Resilience	SR0-SR5	Planned
9	Space Economics and Fleet Finance	EF0-EF5	Planned
10	Space Capstone and Assurance	SC0-SC5	Planned

Volume 7, Lunar and Deep-Space Operations, is next. The series closes at ten with the Space Capstone and Assurance.